

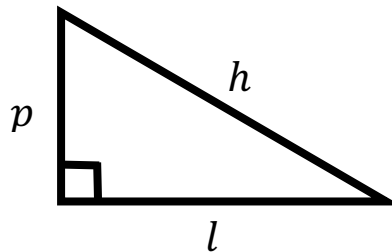
SUSTITUCIÓN TRIGONOMÉTRICA

Propósito: Determinar integrales indefinidas a través de sustituciones trigonométricas.

El **objetivo de las sustituciones trigonométricas** es **eliminar al radical** en el integrando. Esta eliminación se hace utilizando las **identidades pitagóricas**.

$\sin^2(x) + \cos^2(x) = 1$	$\sec^2(x) = \tan^2(x) + 1$	$\tan^2(x) = \sec^2(x) - 1$
$\sin(\alpha) = \frac{co}{h}$	$\cos(\alpha) = \frac{ca}{h}$	$\tan(\alpha) = \frac{co}{ca}$
$\csc(\alpha) = \frac{h}{co}$	$\sec(\alpha) = \frac{h}{ca}$	$\cot(\alpha) = \frac{ca}{co}$

TEOREMA DE PITÁGORAS

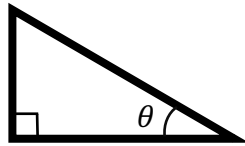


$$p^2 + l^2 = h^2$$

$\sqrt{a^2 - x^2}$	$\sqrt{a^2 + x^2}$	$\sqrt{x^2 - a^2}$
$\sin(\theta) = \frac{x}{a}$	$\tan(\theta) = \frac{x}{a}$	$\sec(\theta) = \frac{x}{a}$
$\sin(\theta) = \frac{x}{a}$	$\tan(\theta) = \frac{x}{a}$	$\sec(\theta) = \frac{x}{a}$

Ejemplo: Determinar

$$\int \frac{dx}{x^2 \sqrt{9 - x^2}}$$



Longitud de la hipotenusa:

Longitud del cateto opuesto al ángulo θ :

$$\sin(\theta) =$$

$$\sin(\theta) =$$

$$= dx$$

$$\int \frac{dx}{x^2 \sqrt{9 - x^2}}$$

$$= \int \frac{3 \cos(\theta) d\theta}{(3 \sin(\theta))^2 \sqrt{9 - (3 \sin(\theta))^2}}$$

$$= \int \frac{3 \cos(\theta) d\theta}{9 \sin^2(\theta) \sqrt{9 - 9 \sin^2(\theta)}}$$

$$= \int \frac{3 \cos(\theta) d\theta}{9 \sin^2(\theta) \sqrt{9[1 - \sin^2(\theta)]}}$$

$$= \int \frac{3 \cos(\theta) d\theta}{9 \sin^2(\theta) \sqrt{9} \sqrt{1 - \sin^2(\theta)}}$$

$$= \int \frac{3 \cos(\theta) d\theta}{27 \sin^2(\theta) \sqrt{1 - \sin^2(\theta)}}$$

$$\sin^2(x) + \cos^2(x) = 1, \quad \cos^2(x) = 1 - \sin^2(x)$$

$$= \int \frac{3 \cos(\theta) d\theta}{27 \sin^2(\theta) \sqrt{\cos^2(\theta)}}$$

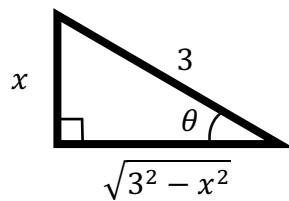
$$= \int \frac{3 \cos(\theta) d\theta}{27 \sin^2(\theta) \cos(\theta)}$$

$$= \frac{1}{9} \int \frac{d\theta}{\sin^2(\theta)}$$

$$= \frac{1}{9} \int \csc^2(\theta) d\theta$$

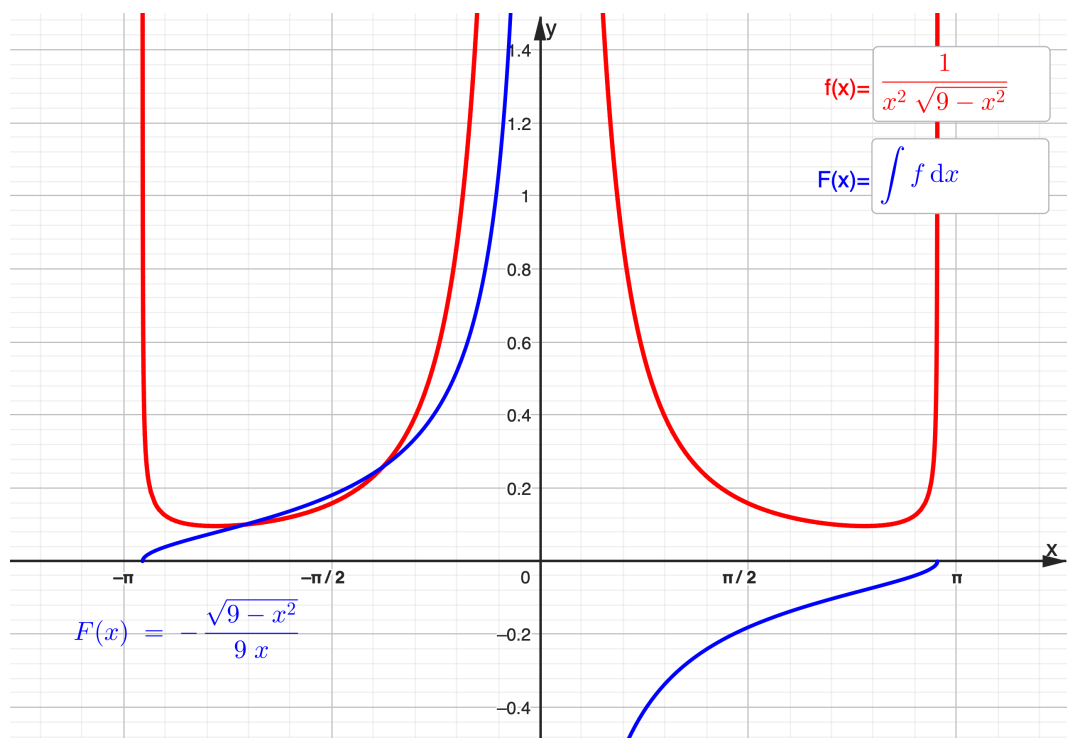
$$= -\frac{1}{9} \cot(\theta) + C$$

$$\cot(\theta) = \frac{ca}{co}$$



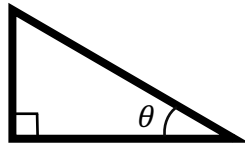
$$\cot(\theta) = \frac{\sqrt{9 - x^2}}{x}$$

$$= -\frac{1}{9} \left(\frac{\sqrt{9 - x^2}}{x} \right) + C$$



Ejemplo: Determinar

$$\int \frac{dx}{\sqrt{4x^2 + 1}}$$



Longitud del cateto adyacente al ángulo θ :

Longitud del cateto opuesto al ángulo θ :

$$\tan(\theta) = 2x$$

$$\frac{\tan(\theta)}{2} =$$

$$= dx$$

$$\int \frac{dx}{\sqrt{4x^2 + 1}}$$

$$= \int \frac{\frac{\sec^2(\theta)}{2}}{\sqrt{4\left(\frac{\tan(\theta)}{2}\right)^2 + 1}} d\theta$$

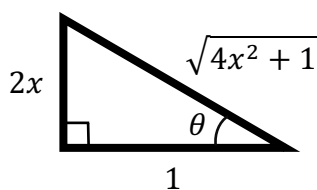
$$= \int \frac{\frac{\sec^2(\theta)}{2}}{\sqrt{4\left(\frac{\tan^2(\theta)}{4}\right) + 1}} d\theta$$

$$= \int \frac{\frac{\sec^2(\theta)}{2}}{\sqrt{\tan^2(\theta) + 1}} d\theta$$

$$\tan^2(x) = \sec^2(x) - 1, \quad \sec^2(x) = \tan^2(x) + 1$$

$$= \int \frac{\frac{\sec^2(\theta)}{2}}{\sqrt{\sec^2(\theta)}} d\theta$$

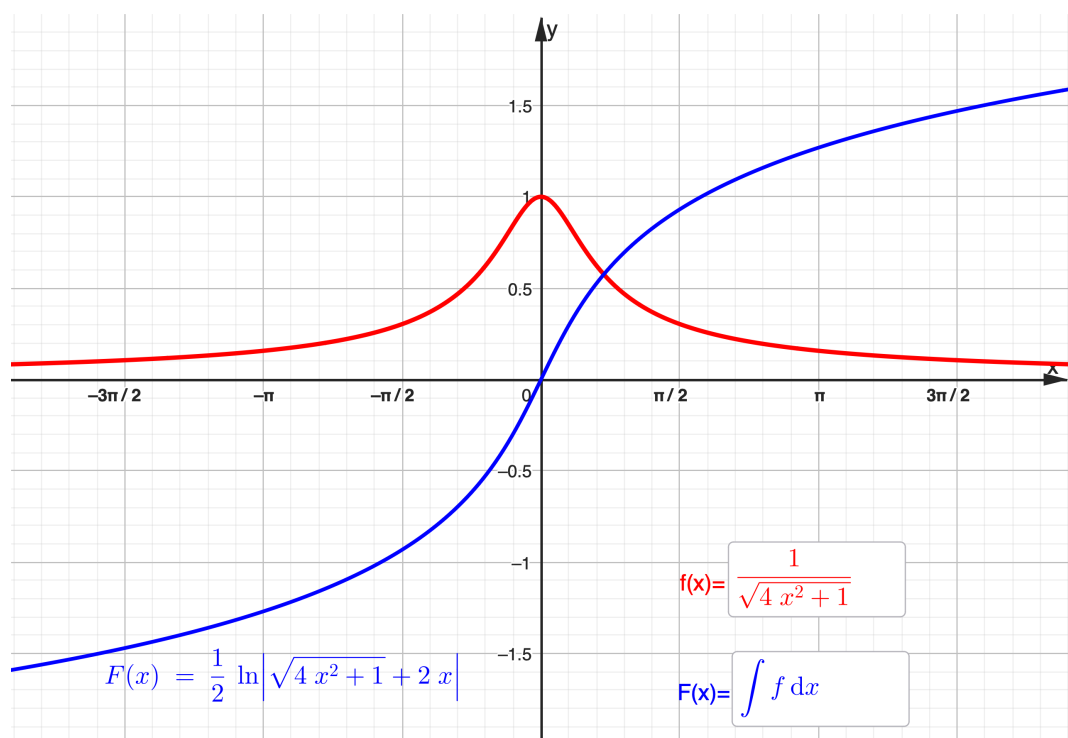
$$\begin{aligned}
 &= \int \frac{\frac{\sec^2(\theta)}{2}}{\sec(\theta)} d\theta \\
 &= \int \frac{\sec^2(\theta)}{2 \sec(\theta)} d\theta \\
 &= \frac{1}{2} \int \sec(\theta) d\theta \\
 &= \frac{1}{2} \ln |\sec(\theta) + \tan(\theta)| + C
 \end{aligned}$$



$$\sec(\theta) = \frac{h}{ca}, \quad \sec(\theta) = \frac{\sqrt{4x^2 + 1}}{1}$$

$$\tan(\theta) = \frac{co}{ca}, \quad \tan(\theta) = \frac{2x}{1}$$

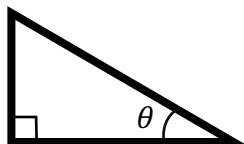
$$= \frac{1}{2} \ln \left| \sqrt{4x^2 + 1} + 2x \right| + C$$



Ejemplo: Determinar

$$\int \frac{dx}{(x^2 - 1)^{3/2}}$$

$$= \int \frac{dx}{\sqrt{(x^2 - 1)^3}}$$



Longitud de la hipotenusa:

Longitud del cateto adyacente al ángulo θ :

$$\sec(\theta) =$$

$$\sec(\theta) =$$

$$= dx$$

$$\int \frac{dx}{\sqrt{(x^2 - 1)^3}}$$

$$= \int \frac{\sec(\theta) \tan(\theta)}{\sqrt{(\sec^2(\theta) - 1)^3}} d\theta$$

$$\tan^2(x) = \sec^2(x) - 1$$

$$= \int \frac{\sec(\theta) \tan(\theta)}{\sqrt{(\tan^2(\theta))^3}} d\theta$$

$$= \int \frac{\sec(\theta) \tan(\theta)}{\sqrt{(\tan^3(\theta))^2}} d\theta$$

$$= \int \frac{\sec(\theta) \tan(\theta)}{\tan^3(\theta)} d\theta$$

$$= \int \frac{\sec(\theta)}{\tan^2(\theta)} d\theta$$

$$\sec(\theta) = \frac{1}{\cos(\theta)}, \quad \tan^2(\theta) = \frac{\sin^2(\theta)}{\cos^2(\theta)}$$

$$= \int \frac{\frac{1}{\cos(\theta)}}{\frac{\sin^2(\theta)}{\cos^2(\theta)}} d\theta$$

$$= \int \frac{\cos^2(\theta)}{\cos(\theta) \sin^2(\theta)} d\theta$$

$$= \int \frac{\cos(\theta)}{\sin^2(\theta)} d\theta$$

$$u = \sin(\theta), \quad du = \cos(\theta) d\theta$$

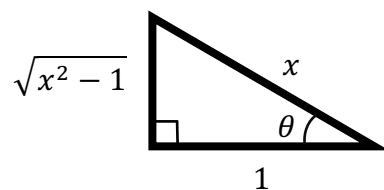
$$= \int \frac{1}{u^2} du$$

$$= \int u^{-2} du$$

$$= -u^{-1} + C$$

$$= -(\sin(\theta))^{-1} + C$$

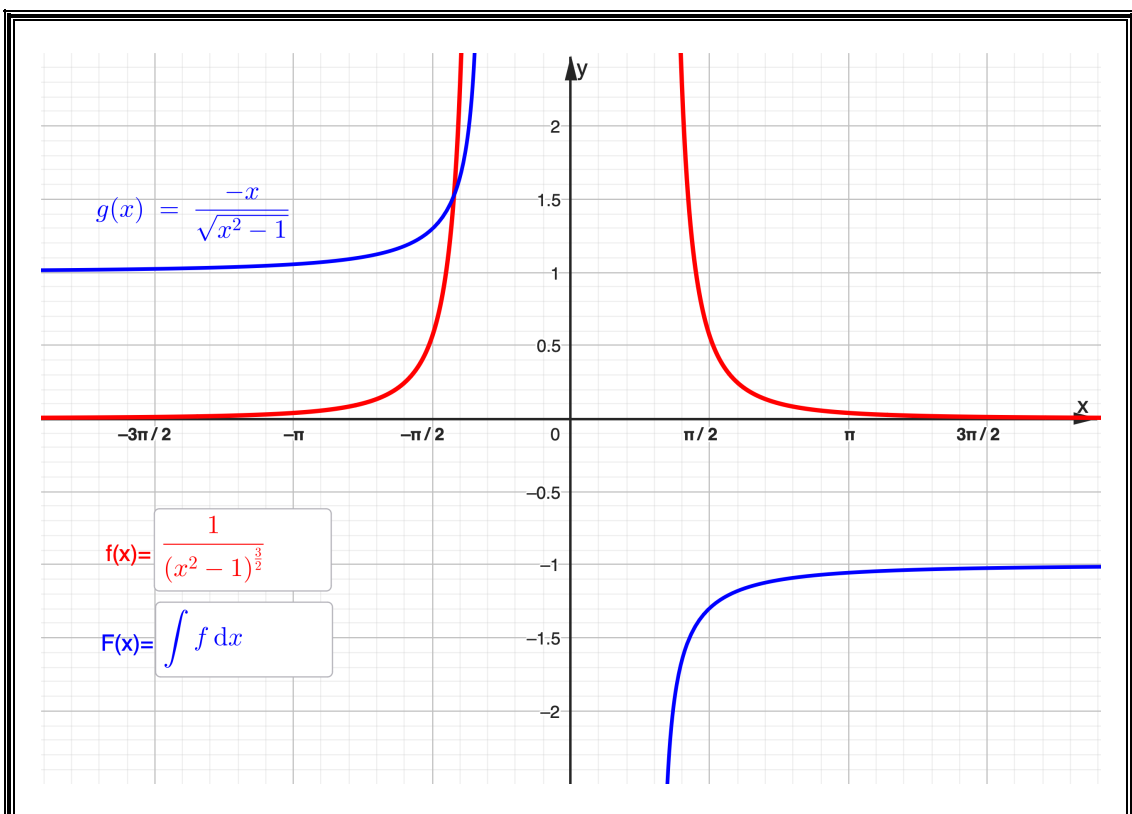
$$= -\frac{1}{\sin(\theta)} + C$$



$$\sin(\theta) = \frac{\text{opposite}}{\text{hypotenuse}}, \quad \sin(\theta) = \frac{\sqrt{x^2 - 1}}{x}$$

$$= -\frac{1}{\frac{\sqrt{x^2 - 1}}{x}} + C$$

$$= -\frac{x}{\sqrt{x^2 - 1}} + C$$



DERIVADAS DE LAS FUNCIONES TRIGONOMÉTRICAS

$\frac{d}{dx} [\sin(x)] = \cos(x)$	$\frac{d}{dx} [\cos(x)] = -\sin(x)$	$\frac{d}{dx} [\tan(x)] = \sec^2(x)$
$\frac{d}{dx} [\cot(x)] = -\csc^2(x)$	$\frac{d}{dx} [\sec(x)] = \sec(x) \tan(x)$	$\frac{d}{dx} [\csc(x)] = -\csc(x) \cot(x)$

ALGUNAS INTEGRALES

$\int \sin(x) \, dx = -\cos(x) + C$	$\int \cos(x) \, dx = \sin(x) + C$
$\int \tan(x) \, dx = \ln \sec(x) + C$	$\int \cot(x) \, dx = \ln \sin(x) + C$
$\int \sec(x) \, dx = \ln \sec(x) + \tan(x) + C$	$\int \csc(x) \, dx = \ln \csc(x) - \cot(x) + C$
$\int \sec^2(x) \, dx = \tan(x) + C$	$\int \csc^2(x) \, dx = -\cot(x) + C$
$\int \sec(x) \tan(x) \, dx = \sec(x) + C$	$\int \csc(x) \cot(x) \, dx = -\csc(x) + C$

EJERCICIOS

Evaluar las siguientes integrales y verificar la solución derivando:

a.	$\int \csc(x) \, dx$
b.	$\int \frac{1}{x\sqrt{4x^2 + 9}} \, dx$
c.	$\int \frac{\sqrt{x^2 - 3}}{x} \, dx$
d.	$\int -\frac{x}{\sqrt{x^2 - 1}} \, dx$
e.	$\int \frac{\sqrt{x^2 - 2x - 3}}{x - 1} \, dx$
f.	$\int \sqrt{9 - x^2} \, dx$
g.	$\int x\sqrt{9 - x^2} \, dx$
h.	$\int \frac{x}{\sqrt{x^2 - 4}} \, dx$
i.	$\int \frac{x^4}{\sqrt{1 - x^2}} \, dx$

SOLUCIONES DE LOS EJERCICIOS

Evaluar las siguientes integrales y verificar la solución derivando:

a.	$\ln \csc(x) - \cot(x) + C$
b.	$-\frac{1}{3} \ln \left \frac{\sqrt{4x^2 + 9}}{2x} - \frac{3}{2x} \right + C$
c.	$\sqrt{3} \left(\frac{\sqrt{x^2 - 3}}{\sqrt{3}} - \sec^{-1} \left(\frac{x}{\sqrt{3}} \right) \right) + C$

d.	$-\sqrt{x^2 - 1} + C$
e.	$4\left(\frac{2}{\sqrt{(x-1)^2 - 4}}\right) + \tan^{-1}\left(\frac{2}{\sqrt{(x-1)^2 - 4}}\right) + C$
f.	$\frac{9}{2}\sin^{-1}\left(\frac{x}{3}\right) + \frac{x\sqrt{9-x^2}}{2} + C$
g.	$= -\frac{\sqrt{(9-x^2)^3}}{3} + C$
h.	$\sqrt{x^2 - 4} + C$
i.	$\frac{3}{8}\sin^{-1}(x) - \frac{3x\sqrt{1-x^2}}{8} - \frac{x^3\sqrt{1-x^2}}{4} + C$